

Making BIOMOD2 executable and measurable on HPC

Species Distribution Models in R · workflow, resources and experiments

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Question. How can an R/BIOMOD2 workflow be made reproducible, executable and measurable on an HPC system without confusing computational optimisations with scientific results?

- A large presence dataset and raster projections create a substantial memory and I/O load.
- The workflow combines five algorithms, ensemble models, current climate and eight future scenarios.
- The experimental comparison focuses on raster representation and number of workers.

Input data

2,583,359 presence records
167 species
1 km² resolution

Models

GLM · GBM · ANN
FDA · MAXNET
two ensembles

Execution

R + BIOMOD2
Terra + Singularity
Slurm job arrays

calibration → projection → isolated outputs → validation

```
#!/bin/bash
#SBATCH --array=0-166
#SBATCH --cpus-per-task=8
#SBATCH --mem=300G

Rscript run_species.R --species-index $SLURM_ARRAY_TASK_ID
```

- One species per array task isolates outputs and failures.
- Resources are explicit and measurable.
- A completed job is not automatically a validated scientific result.

Experiment	Observed result	Status
Storage FALSE/FALSE	about 261 GiB MaxRSS	completed
Storage FALSE/TRUE	clamping-mask failure	to investigate
Workers 4 / 6 / 8	completed for <code>Achillea.atrata</code>	validated
Workers 32	out of memory	failed

What the template is for

Long technical talk

- workflow and design
- resource model
- experimental matrix
- detailed plots and limitations

Short defence

- one problem slide
- one pipeline slide
- two result slides
- one conclusion slide

Rule: one message per slide, one evidence label per result.

- Reproducibility requires isolating species, scenarios and execution resources.
- RAM, I/O and parallelism are coupled; more workers do not guarantee better throughput.
- Results must distinguish completed runs, validated outputs and scientific conclusions.

Questions?